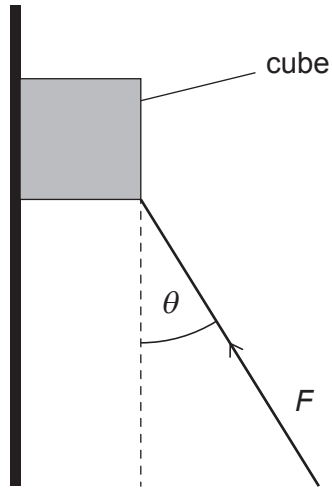


PAPER 1 / QUESTION 47

29 April 2025 / TZ1 (file label) / HL Paper 1A / original question 3 / 2225-6707

47. A force acts on a cube of mass m that accelerates upwards along a vertical frictionless surface.

The magnitude of the force is F and it acts at θ to the vertical.



What is the magnitude of the acceleration of the cube?

- A. $\frac{(F \cos \theta - mg)}{m}$
- B. $\frac{(F \sin \theta - mg)}{m}$
- C. $\frac{(F \cos \theta - g)}{m}$
- D. $\frac{(F \sin \theta - g)}{m}$